

Homework of Chapter 2

2.1 a Demonstrate by means of truth tables the validity of the following identities:

(a) DeMorgan's theorem for three variables: $\overline{XYZ} = \overline{X} + \overline{Y} + \overline{Z}$

(b) The second distributive law: $X + YZ = (X + Y)(X + Z)$

(c) $\overline{XY} + \overline{YZ} + \overline{XZ} = X\overline{Y} + Y\overline{Z} + \overline{X}Z$

Answer

(a) $\overline{XYZ} = \overline{X} + \overline{Y} + \overline{Z}$

X	Y	Z	$\overline{XYZ}$	$\overline{X} + \overline{Y} + \overline{Z}$
0	0	0	1	1
1	0	0	1	1
0	1	0	1	1
0	0	1	1	1
1	1	0	1	1
1	0	1	1	1
0	1	1	1	1
1	1	1	0	0

(b) $X + YZ = (X + Y)(X + Z)$

X	Y	Z	$X + YZ$	$(X + Y)(X + Z)$
0	0	0	0	0
1	0	0	1	1
0	1	0	0	0
0	0	1	0	0

X	Y	Z	$X+YZ$	$(X+Y)(X+Z)$
1	1	0	1	1
1	0	1	1	1
0	1	1	0	0
1	1	1	1	1

(c) $\overline{X}Y + \overline{Y}Z + X\overline{Z} = X\overline{Y} + Y\overline{Z} + \overline{X}Z$

X	Y	Z	$\overline{X}Y + \overline{Y}Z + X\overline{Z}$	$X\overline{Y} + Y\overline{Z} + \overline{X}Z$
0	0	0	0	0
1	0	0	1	1
0	1	0	1	1
0	0	1	1	1
1	1	0	1	1
1	0	1	1	1
0	1	1	1	1
1	1	1	0	0

2.2 a/c *Prove the identity of each of the following Boolean equations, using algebraic manipulation:

(a) $\overline{X}\overline{Y} + \overline{X}Y + XY = \overline{X} + Y$

(c) $Y + \overline{X}Z + X\overline{Y} = X + Y + Z$

Answer

(a)

$$\begin{aligned}
& \overline{X} \overline{Y} + \overline{X} Y + X Y \\
&= \overline{X} \overline{Y} + \overline{X} Y + \overline{X} Y + X Y \\
&= \overline{X} (\overline{Y} + Y) + Y (\overline{X} + X) \\
&= \overline{X} + Y
\end{aligned}$$

(b)

$$\begin{aligned}
& Y + \overline{X} Z + X \overline{Y} \\
&= Y (\overline{X} + X) + \overline{X} Z + X \overline{Y} \\
&= X (\overline{Y} + Y) + \overline{X} Z + \overline{X} Y \\
&= X + \overline{X} Z + \overline{X} Y \\
&= X (1 + Z) + \overline{X} Z + \overline{X} Y \\
&= Z + X + \overline{X} Y \\
&= X + Y + Z
\end{aligned}$$

2.3 a/c Prove the identity of each of the following Boolean equations, using algebraic manipulation:

(a) $AB\overline{C} + B\overline{C}\overline{D} + BC + \overline{C}D = B + \overline{C}D$

(c) $A\overline{D} + \overline{A}B + \overline{C}D + \overline{B}C = (\overline{A} + \overline{B} + \overline{C} + \overline{D})(A + B + C + D)$

Answer

(a)

$$\begin{aligned}
& AB\overline{C} + B\overline{C}\overline{D} + BC + \overline{C}D \\
&= AB\overline{C} + BC(A + 1) + B\overline{C}\overline{D} + \overline{C}D(B + 1) \\
&= AB(\overline{C} + C) + BC + B\overline{C}(\overline{D} + D) + \overline{C}D \\
&= AB + BC + B\overline{C} + \overline{C}D \\
&= B(A + C + \overline{C}) + \overline{C}D \\
&= B + \overline{C}D
\end{aligned}$$

(c)

辅助证明

$$\begin{aligned}
& \overline{A}B + \overline{B}C + \overline{A}C \\
&= \overline{A}B + \overline{B}C + \overline{A}C(B + \overline{B}) \\
&= \overline{A}B(1 + C) + \overline{B}C(1 + \overline{A}) \\
&= \overline{A}B + \overline{B}C
\end{aligned}$$

根据上面的辅助证明，我们得到了布尔代数中的一致性定理推论，即任意满足形式 $XY + \overline{X}Z + YZ$ 的三项式中，多余的 YZ 可以被完全吸收。

现在我们来处理题目的等式右侧，将它全部展开：

$$\begin{aligned}
& (\overline{A} + \overline{B} + \overline{C} + \overline{D})(A + B + C + D) \\
&= 0 + \overline{A}B + \overline{A}C + \overline{A}D \\
&+ \overline{B}A + 0 + \overline{B}C + \overline{B}D \\
&+ \overline{C}A + \overline{C}B + 0 + \overline{C}D \\
&+ \overline{D}A + \overline{D}B + \overline{D}C + 0
\end{aligned}$$

展开后一共有 12 项不为 0 的项，我们要证明它们最终等于左侧的 4 个目标项： $\overline{A}D + \overline{A}B + \overline{C}D + \overline{B}C$ 。因此我们要把剩下的 8 项全都通过辅助证明的方法“吸收”掉。

为了更直观得得到左侧目标，我们以目标项基点作为吸收主力：

第一步：核心目标项直接进行初步清理（以目标 $\overline{A}D, \overline{A}B, \overline{C}D, \overline{B}C$ 作为发起点）

1. $\overline{A}D + \overline{C}D + \overline{A}C = \overline{A}D + \overline{C}D$ ，吸收项 $\overline{C}A$ ($\overline{A}C$)。
2. $\overline{A}B + \overline{B}C + \overline{A}C = \overline{A}B + \overline{B}C$ ，吸收项 $\overline{A}C$ 。
3. $\overline{B}C + \overline{C}D + \overline{B}D = \overline{B}C + \overline{C}D$ ，吸收项 $\overline{B}D$ 。
4. $\overline{A}B + \overline{A}D + \overline{B}D = \overline{A}B + \overline{A}D$ ，吸收项 $\overline{B}D$ ($\overline{B}D$)。

此时 12 个项被吸收了 4 个，还剩下 8 个。这 8 项中包含我们的 4 个终极目标项 ($\overline{A}D, \overline{A}B, \overline{B}C, \overline{C}D$)，以及 4 个残余项 ($\overline{A}D, \overline{A}B, \overline{B}C, \overline{C}D$)。

第二步：利用池子中剩余的其他项作为中间桥梁，继续吸收剩余 4 项（基于同样的辅助证明定律）

在未被完全剥离的第一轮展开组合中，我们依然可以运用一致性定理找出消去残余项的关系：

5. 因为 $\overline{A}C + \overline{C}D + \overline{A}D = \overline{A}C + \overline{C}D$ ，吸收项 $\overline{A}D$ 。
6. 因为 $\overline{A}C + \overline{B}C + \overline{A}B = \overline{A}C + \overline{B}C$ ，吸收项 $\overline{A}B$ ($\overline{A}B$)。
7. 因为 $\overline{B}D + \overline{C}D + \overline{B}C = \overline{B}D + \overline{C}D$ ，吸收项 $\overline{C}B$ ($\overline{B}C$)。
8. 因为 $\overline{A}C + \overline{A}D + \overline{C}D = \overline{A}C + \overline{A}D$ ，吸收项 $\overline{C}D$ ($\overline{C}D$)。

经过这两轮纯代数吸收操作，原式中除了目标项外的所有 8 个多余项（4 个过渡项和 4 个残留项）都被利用一致性定理交叉消去了。

因此：

$$= A\overline{D} + \overline{A}B + \overline{C}D + \overline{B}C$$

2.6 b/d Simplify the following Boolean expressions to expressions containing a minimum number of literals:

(b) $(\overline{A + B + C}) \cdot \overline{ABC}$

(d) $\overline{A} \overline{B} D + \overline{A} \overline{C} D + BD$

Answer

(b)

$$\begin{aligned} & (\overline{A + B + C}) \cdot \overline{ABC} \\ &= (\overline{A} \overline{B} \overline{C}) \cdot (\overline{A} + \overline{B} + \overline{C}) \\ &= \overline{A} \overline{B} \overline{C} \cdot \overline{A} + \overline{A} \overline{B} \overline{C} \cdot \overline{B} + \overline{A} \overline{B} \overline{C} \cdot \overline{C} \\ &= \overline{A} \overline{B} \overline{C} + \overline{A} \overline{B} \overline{C} + \overline{A} \overline{B} \overline{C} \\ &= \overline{A} \overline{B} \overline{C} \end{aligned}$$

(d)

$$\begin{aligned} & \overline{A} \overline{B} D + \overline{A} \overline{C} D + BD \\ &= D(\overline{A} \overline{B} + \overline{A} \overline{C} + B) \\ &= D(B + \overline{B} \overline{A} + \overline{A} \overline{C}) \\ &= D(B + \overline{A} + \overline{A} \overline{C}) \\ &= D(B + \overline{A}(1 + \overline{C})) \\ &= D(\overline{A} + B) \end{aligned}$$

2.10 a/c *Obtain the truth table of the following functions, and express each function in sum-of-minterms and product-of-maxterms form:

(a) $(XY + Z)(Y + XZ)$

(c) $WX\overline{Y} + WX\overline{Z} + WXZ + Y\overline{Z}$

Answer

(a) 先化简出判断条件: $F = (XY + Z)(Y + XZ) = XY + XYZ + YZ + XZ = XY + YZ + XZ$

\$X\$	\$Y\$	\$Z\$	\$F\$
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	1

- **Sum-of-minterms:** $F(X, Y, Z) = \sum m(3, 5, 6, 7)$
- **Product-of-maxterms:** $F(X, Y, Z) = \prod M(0, 1, 2, 4)$

(c) 先化简: $F = WX\bar{Y} + WX\bar{Z} + WXZ + Y\bar{Z} = WX\bar{Y} + WX(\bar{Z} + Z) + Y\bar{Z} = WX(1 + \bar{Y}) + Y\bar{Z} = WX + Y\bar{Z}$

\$W\$	\$X\$	\$Y\$	\$Z\$	\$F\$
0	0	0	0	0
0	0	0	1	0
0	0	1	0	1
0	0	1	1	0
0	1	0	0	0
0	1	0	1	0
0	1	1	0	1
0	1	1	1	0
1	0	0	0	0

\$W\$	\$X\$	\$Y\$	\$Z\$	\$F\$
1	0	0	1	0
1	0	1	0	1
1	0	1	1	0
1	1	0	0	1
1	1	0	1	1
1	1	1	0	1
1	1	1	1	1

- **Sum-of-minterms:** $F(W, X, Y, Z) = \sum m(2, 6, 10, 12, 13, 14, 15)$
- **Product-of-maxterms:** $F(W, X, Y, Z) = \prod M(0, 1, 3, 4, 5, 7, 8, 9, 11)$

2.11 a/c/d

For the Boolean functions E and F, as given in the following truth table:

- List the minterms and maxterms of each function.
- List the minterms of $E + F$ and $E \cdot F$
- Express E and F in sum-of-minterms algebraic form.

X	Y	Z	E	F
0	0	0	0	1
0	0	1	1	0
0	1	0	1	1
0	1	1	0	0
1	0	0	1	1
1	0	1	0	0

X	Y	Z	E	F
1	1	0	1	0
1	1	1	0	1

Answer

(a)

- E 的最小项与最大项: $E = \sum m(1, 2, 4, 6) = \prod M(0, 3, 5, 7)$
- F 的最小项与最大项: $F = \sum m(0, 2, 4, 7) = \prod M(1, 3, 5, 6)$

(c)

- $E + F$ 的最小项: $\sum m(0, 1, 2, 4, 6, 7)$
- $E \cdot F$ 的最小项: $\sum m(2, 4)$

(d)

- E 的最小项之和代数表达式: $E = \overline{X} \overline{Y} Z + \overline{X} Y \overline{Z} + X \overline{Y} \overline{Z} + X Y \overline{Z}$
- F 的最小项之和代数表达式: $F = \overline{X} \overline{Y} \overline{Z} + \overline{X} Y \overline{Z} + X \overline{Y} \overline{Z} + X Y Z$

2.12 b Convert the following expressions into sum- of- products and product- of-sums forms:

(b) $\overline{X} + X(X + \overline{Y})(Y + \overline{Z})$

Answer

$$\begin{aligned}
 F &= \overline{X} + X(X + \overline{Y})(Y + \overline{Z}) \\
 &= \overline{X} + (X + X\overline{Y})(Y + \overline{Z}) \\
 &= \overline{X} + X(Y + \overline{Z}) \quad (\text{利用吸收律 } X + X\overline{Y} = X) \\
 &= \overline{X} + XY + X\overline{Z} \\
 &= \overline{X} + Y + X\overline{Z} \quad (\text{利用吸收律 } \overline{X} + XY = \overline{X} + Y) \\
 &= \overline{X} + Y + \overline{Z} \quad (\text{利用吸收律 } \overline{X} + X\overline{Z} = \overline{X} + \overline{Z})
 \end{aligned}$$

- 乘积之和 (SOP): $\overline{X} + Y + \overline{Z}$
- 和之乘积 (POS): $\overline{X} + Y + \overline{Z}$

(注: $\overline{X} + Y + \overline{Z}$ 可视为一个独立项, 同时满足最简的 SOP 和 POS 形式)。

2.15 Optimize the following Boolean expressions using a map:

(a) $\overline{X} \overline{Z} + Y \overline{Z} + XYZ$

(b) $\overline{A}B + \overline{B}C + \overline{A} \overline{B} \overline{C}$

(c) $\overline{A} \overline{B} + \overline{A} \overline{C} + \overline{B}C + \overline{A}B \overline{C}$

Answer

(a) 包含最小项 $m(0, 2, 6, 7) \Rightarrow F = \overline{X} \overline{Z} + XY$

(b) 包含最小项 $m(0, 1, 2, 3, 5) \Rightarrow F = \overline{A} + \overline{B}C$

(c) 包含最小项 $m(0, 1, 2, 4, 5, 6) \Rightarrow F = \overline{B} + \overline{C}$

2.17 Optimize the following Boolean functions, using a map:

(a) $F(W, X, Y, Z) = \sum m(0, 1, 2, 4, 7, 8, 10, 12)$

(b) $F(A, B, C, D) = \sum m(1, 4, 5, 6, 10, 11, 12, 13, 15)$

Answer

(a) $F = \overline{X} \overline{Z} + \overline{Y} \overline{Z} + \overline{W} \overline{X} \overline{Y} + \overline{W}XYZ$

(b) $F = \overline{A} \overline{C}D + \overline{A}B \overline{D} + \overline{B}C + \overline{A} \overline{B}C + ACD$

2.19 a Find all the prime implicants for the following Boolean functions, and determine which are essential:

(a) $F(W, X, Y, Z) = \sum m(0, 2, 5, 7, 8, 10, 12, 13, 14, 15)$

Answer

(a)

- 质蕴含项 (Prime Implicants): $\overline{X} \overline{Z}, XZ, WX, W\overline{Z}$
- 核心质蕴含项 (Essential Prime Implicants): $\overline{X} \overline{Z}$ (覆盖了 m_0, m_2) 以及 XZ (覆盖了 m_5, m_7)

2.22 a Optimize the following expressions in (1) sum-of-products and (2) product-of-sums forms:

(a) $\overline{A} \overline{C} + \overline{B}D + \overline{A}CD + ABCD$

Answer

(a) 最小项: $m(1, 3, 7, 8, 9, 11, 12, 13, 15)$, 最大项 (零点): $M(0, 2, 4, 5, 6, 10, 14)$

- **(1) 乘积之和 (SOP):** $F = A\overline{C} + \overline{B}D + CD$
- **(2) 和之乘积 (POS):** $F = (A + D)(\overline{C} + D)(A + \overline{B} + C)$

2.25 b Optimize the following Boolean functions F together with the don't-care conditions d . Find all prime implicants and essential prime implicants, and apply the selection rule.

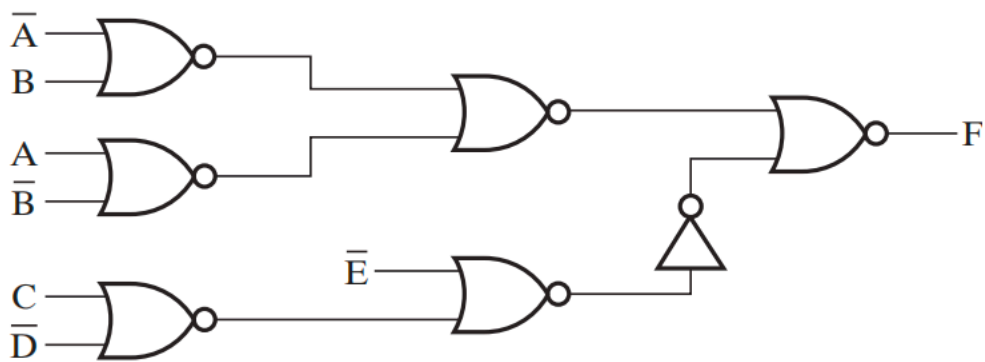
(b) $F(W, X, Y, Z) = \sum m(0, 2, 4, 5, 8, 14, 15), d(W, X, Y, Z) = \sum m(7, 10, 13)$

Answer

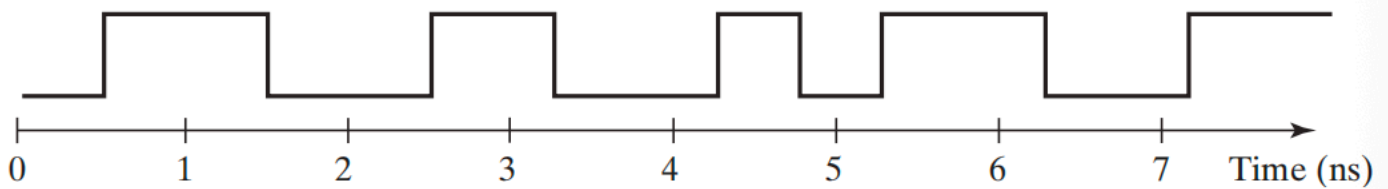
(b)

- **质蕴含项 (Prime Implicants):** $\overline{X}\overline{Z}, XZ, \overline{W}\overline{Y}\overline{Z}, \overline{W}X\overline{Y}, WXY, WY\overline{Z}$
- **核心质蕴含项 (Essential Prime Implicants):** $\overline{X}\overline{Z}$ (唯一覆盖了 m_2 和 m_8)
- **选择规则 (Selection Rule):** 我们首先选取核心质蕴含项 $\overline{X}\overline{Z}$ 。剩余需要覆盖的最小项为 4, 5, 14, 15。与其选择 XZ 造成覆盖重叠并导致额外还需要 2 个独立的质蕴含项拼凑 (总共将形成 4 项组合), 我们直接进行映射划分: $\overline{W}X\overline{Y}$ 可以精确覆盖 4, 5; WXY 可以精确覆盖 14, 15。
- **优化后的最简函数 (Optimized Function):** $F = \overline{X}\overline{Z} + \overline{W}X\overline{Y} + WXY$

2.29 * The NOR gates in Figure 2-39 have propagation delay $t_{pd} = 0.073$ ns and the inverter has a propagation delay $t_{pd} = 0.048$ ns. What is the propagation delay of the longest path through the circuit?



□ **FIGURE 2-39**
Circuit for Problem 2-29



□ **FIGURE 2-40**
Waveform for Problem 2-30

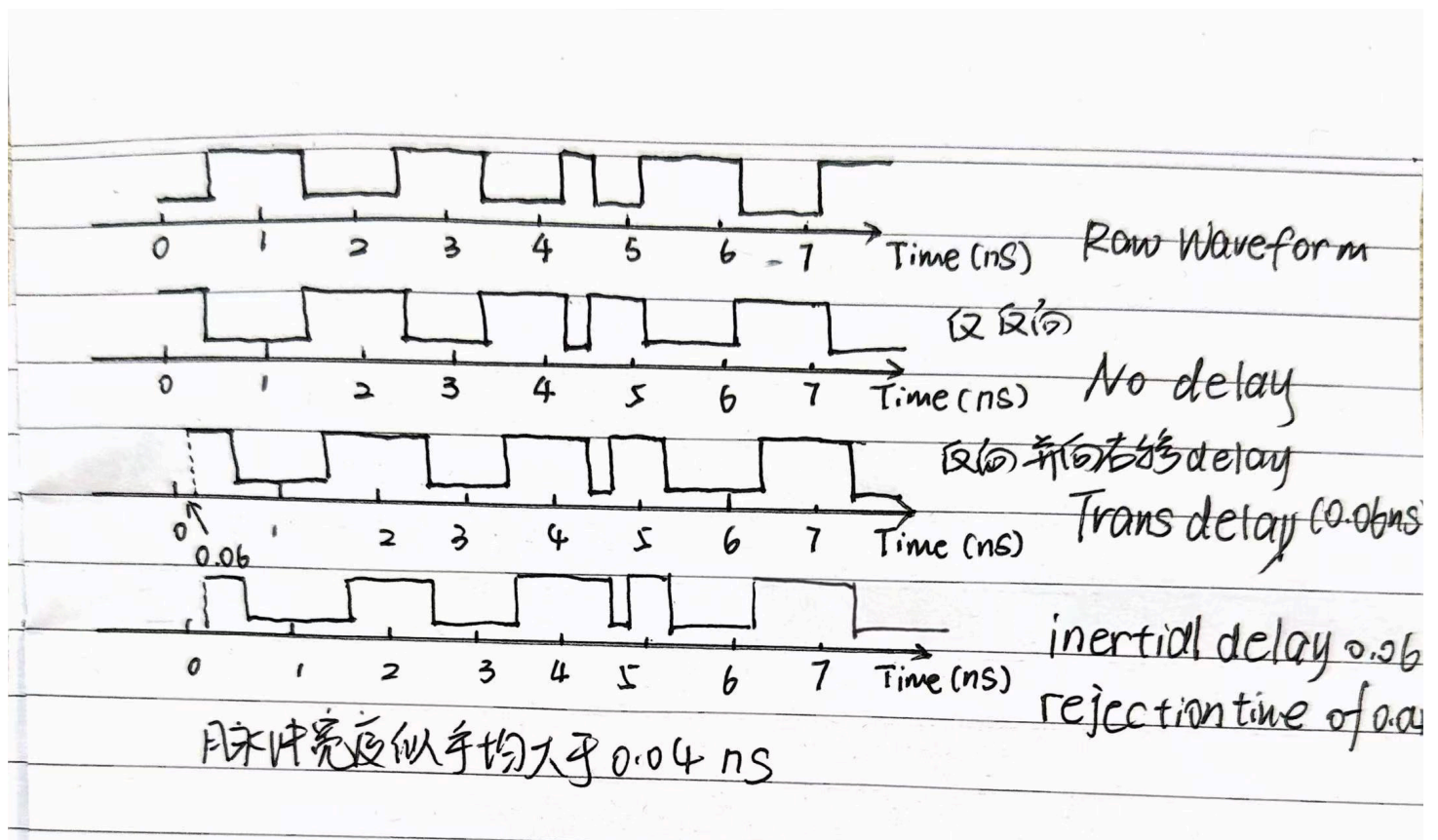
Answer

或非门的延迟为0.073ns,反相器的延迟为0.048ns

则从C口的信号的延迟最长 $0.073 * 2 + 0.048 = 0.194\text{ns}$

2.30 The waveform in Figure 2-40 is applied to an inverter. Find the output of the inverter, assuming that

- (a) It has no delay.
- (b) It has a transport delay of 0.06 ns.
- (c) It has an inertial delay of 0.06 ns with a rejection time of 0.04 ns



2.31 Assume that t_{pd} is the average of t_{PHL} and t_{PLH} . Find the delay from each input to the output in Figure 2-41 by

(a) Finding t_{PHL} and t_{PLH} for each path, assuming $t_{PHL} = 0.20$ ns and $t_{PLH} = 0.36$ ns for each gate. From these values, find t_{pd} for each path.

(b) Using $t_{pd} = 0.28$ ns for each gate.

(c) Compare your answers from parts (a) and (b) and discuss any differences.

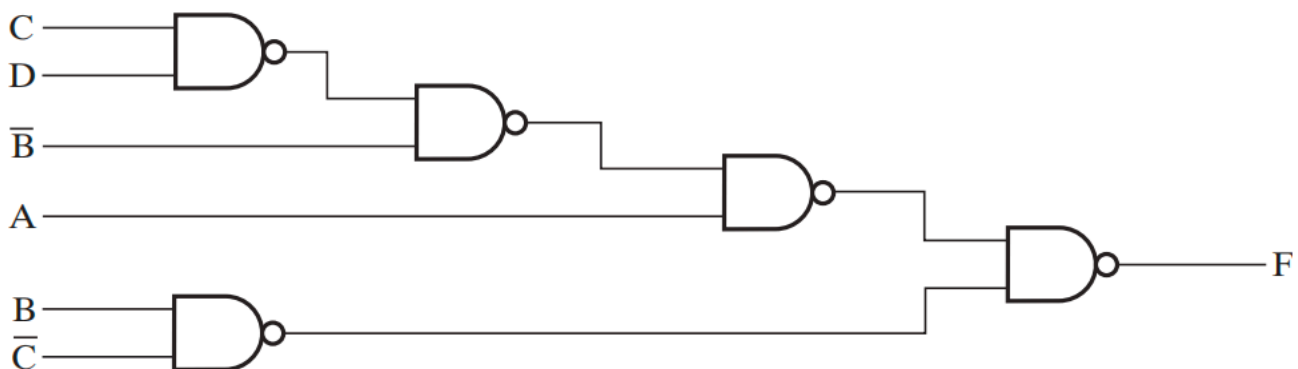


FIGURE 2-41
Circuit for Problem 2-31

Answer

(a)
从高到低是PHL,从低到高是PLH，但是这道题里面的门都是与非门，则直接根据门的数量来计算

Type	SC	SD	$\overline{B}$	A	B	$\overline{C}$
t_{PHL}	0.80 ns	0.80 ns	0.60 ns	0.40 ns	0.40 ns	0.40 ns
t_{PLH}	1.44 ns	1.44 ns	1.08 ns	0.72 ns	0.72 ns	0.72 ns
t_{pd}	1.12 ns	1.12 ns	0.84 ns	0.56 ns	0.56 ns	0.56 ns

- (b) 平均延迟的结果和上一小问的结果完全一样
- (c) 第二小问给出的每个门的平均延迟和第一小问中给出的 t_{PHL} 和 t_{PLH} 的平均值完全一样，所以两者的结果完全一样。但是第一小问的数据可以计算出每条路径的高到低和低到高的延迟，而第二小问只能得到平均值，无法区分不同路径的延迟差异。